

BazarCycle BD

Don t Dump It. Cycle It.

Turning Market Waste into Local Resources

Document	Project Proposal
Competition	Sustainable Project Competition 2026
Institution	Daffodil International University (DIU), Bangladesh
Team	Team PROTON Department of Computer Science and Engineering
Project status	Completed / implemented working prototype (web platform)
SDG focus	SDG 11 SDG 12 SDG 13 (primary) SDG 8 (secondary)

Photograph: vegetable-market waste being cleared from a market lane. Illustrative photograph supplied by the project team. It does not represent measured BazarCycle BD impact.

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2 Executive Summary

BazarCycle BD is a web-based waste-to-resource management platform designed for local markets in Bangladesh. Markets produce vegetable, fruit, fish, plastic, paper/cardboard and other waste every day. When these materials are mixed and disposed of together, recoverable resources are lost and nobody has clear visibility of what was generated, where it went, or whether it was recovered.

The platform gives a market a digital pathway: **Waste Generation Categorization Resource Pathway Collector Recovery Impact Measurement**. A market manager registers waste by category and quantity; a deterministic, rule-based engine recommends a resource pathway; a collector accepts and completes the pickup; and the system updates impact records and a project-defined Sustainability Score.

BazarCycle BD is a **completed, implemented working prototype**. It has not been deployed at scale and is not claimed to be in use by real markets. All resource values and CO_2e figures in this document are **Project Estimates** for demonstration, not independently verified measurements.

What is the problem?	What does BazarCycle BD do differently?	How does it contribute to sustainability?
Market waste is mixed, untracked and hard to coordinate, so recoverable organic and recyclable materials are treated only as disposal burden.	It connects managers, collectors and admins in one workflow with explainable rule-based recommendations, a protected pickup process and measurable impact records.	It supports segregation, composting and recycling (SDG 11, 12, 13) and makes recovery visible through Project Estimate indicators.

3 Background



A busy vegetable market. Markets like this generate organic waste daily. Illustrative photograph supplied by the project team. It does not represent measured BazarCycle BD impact.

Local markets (bazars) are central to urban food supply in Bangladesh. They also concentrate large volumes of **organic waste** such as vegetable trimmings, spoiled fruit and fish waste, together with **recyclable materials** such as plastic and paper/cardboard packaging.

In a circular-economy view, these materials are not only waste. Organic waste can be directed to composting or fertilizer pathways, and plastic and paper can be directed to recycling. Realising this requires three things at market level: correct categorization, a reliable link to someone who can collect the material, and a record of what was recovered.

BazarCycle BD focuses on that coordination and monitoring layer. It does not replace physical composting or recycling facilities; it provides the digital workflow that connects a market to a collector and records the outcome.

4 Problem Statement

Local markets generate vegetable, fruit, fish, plastic, paper/cardboard and other waste every day. When waste is mixed together and disposed of without proper segregation:

Consequence	Why it matters
Recoverable materials are lost	Resources that could be redirected are treated as a single disposal stream.
Organic waste becomes a disposal burden	Vegetable, fruit and fish waste is handled as refuse instead of being routed to composting or fertilizer pathways.
Recyclables are harder to recover	Plastic and paper mixed with wet organic waste are harder to separate and recover.
Collection is difficult to track	There is no shared record of who requested, accepted or completed a pickup.
Limited visibility for markets	Managers lack data on how much of each waste type they generate.
Sustainability is hard to measure	Without records, recovery and improvement cannot be quantified.

The gap BazarCycle BD addresses: a digital coordination and monitoring gap between waste generation at the market and resource recovery. This proposal does not cite national waste statistics; the problem is described from the project s own analysis.

5 Proposed Solution

BazarCycle BD connects five elements into one workflow:

Element	Role in the platform
1. Market Managers	Register waste by category and quantity.
2. Rule-Based Recommendation Engine	Recommends an appropriate resource pathway and estimates resource value.
3. Collectors	View available pickup requests, accept them and mark collections as completed.
4. Impact Monitoring	Tracks waste recovered, composted and recycled, estimated resource value, estimated CO_2e avoided (Project Estimate) and collection activity.
5. Sustainability Score	A project-defined score from 0 to 100 summarising a market s performance.

Design principles. The engine is deterministic and explainable: each recommendation shows the rule that produced it. The platform uses no machine learning, no deep learning and no paid or external AI service. Impact and score values come from records stored in the database rather than static mock-ups.



Composting in concrete rings: an example of a recovery pathway for organic waste. Illustrative photograph supplied by the project team. It does not represent measured BazarCycle BD impact.

6 Objectives

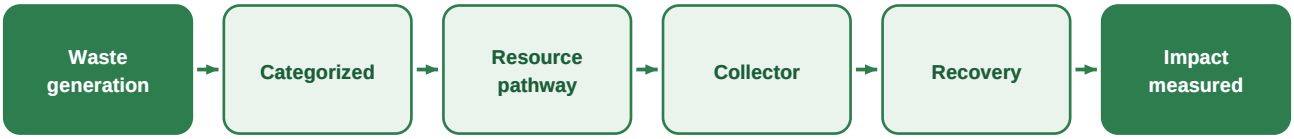
- Provide a single digital workflow from waste registration to measured recovery.
- Encourage **waste segregation** by asking managers to register waste by category.
- Give each waste category an **explainable resource pathway** through deterministic rules.
- Connect markets with collectors through a **transparent, protected pickup process**.
- Measure recovered, composted and recycled waste, estimated resource value and estimated CO_2e (Project Estimate).
- Summarise performance in a **project-defined Sustainability Score** (0 100), not an official certification.
- Demonstrate a working prototype that supports SDG 11, SDG 12 and SDG 13, with SDG 8 as a secondary link.

7 Target Users

User group	Need	How BazarCycle BD helps
Market managers	Organise waste handling and know what the market generates.	Waste registration, pathway recommendations, pickup requests, dashboard and score.
Collectors	Find collection requests and keep a reliable record of work done.	Available-pickup list, acceptance, completion updates and collection history.
Platform administrators	Oversee users, markets and activity.	User and market management, waste registry, pickup tracking, reports.

8 System Workflow

The core journey followed by every waste record:



Stage	What happens
Waste Generation	The market produces vegetable, fruit, fish, plastic, paper/cardboard or other waste.
Categorization	The manager registers the waste with a category and quantity in kg.
Resource Pathway	The rule-based engine recommends a pathway and calculates the estimated resource value.
Collector	The manager requests a pickup; a collector accepts it from the available list.
Recovery	The collector collects the waste and marks the pickup as completed.
Impact Measurement	The system creates impact records and updates the dashboard and Sustainability Score.

9 Waste-to-Resource Recommendation Engine

The recommendation engine is **deterministic and rule-based**. The same category always produces the same pathway, so a manager, collector or judge can see exactly why a result was given.

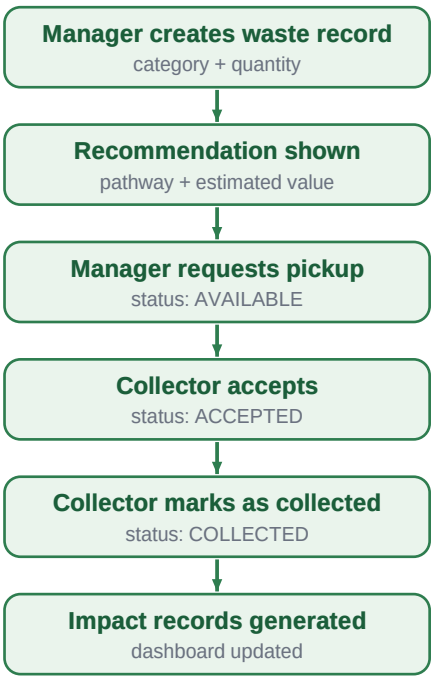
Waste category	Recommended pathway
Vegetable Waste	Composting
Fruit Waste	Composting
Fish Waste	Organic / Fertilizer Pathway
Plastic	Recycling
Paper / Cardboard	Paper Recycling
Other	Responsible Disposal

Estimated Resource Value = Quantity (kg) Estimated Value per kg. Example: 150 kg 15/kg = **2,250**. Values are project estimates for demonstration purposes and may vary by location, quality and market conditions. They are not official market prices.

Example output Category: Vegetable Waste Quantity: 150 kg Recommended pathway: Composting Estimated Resource Value: 2,250 Explanation: vegetable waste can be redirected toward composting instead of mixed disposal.

Requests with a quantity of zero or below, a missing category or a missing market are rejected by backend validation.

10 Pickup & Collection Workflow



Statuses

- **AVAILABLE** requested, waiting for a collector.
- **ACCEPTED** one collector has accepted it.
- **COLLECTED** collection completed.
- **CANCELLED** request withdrawn.

Concurrency protection

A pickup must be accepted by only one collector. The backend validates the pickup state and uses transaction handling so that two collectors cannot accept the same request.

Role control

Only managers create requests; only collectors accept and collect; admins monitor all pickups.

11 Sustainability Score

The **Bazar Sustainability Score** summarises a market s performance from 0 to 100 using four weighted components. It is calculated from records in the database.



Component	Weight
Waste Segregation	25%
Waste Recovery	30%
Recycling & Composting	20%
Collection Efficiency	25%

Score range	Label
0 39	Needs Improvement
40 59	Developing
60 79	Good
80 100	Excellent

Important: this is a **project-defined sustainability metric and is not an official certification.**

12 Impact Measurement

When a collector marks a pickup as collected, the system creates an impact record linked to the market and the waste record. The dashboard aggregates these records.

Indicator	Meaning
Waste registered	Total quantity (kg) registered by the market.
Waste recovered	Quantity (kg) from completed collections.
Waste composted	Recovered quantity that followed the composting pathway.
Waste recycled	Recovered quantity that followed a recycling pathway.
Estimated resource value	Quantity estimated value per kg ( ). **Project Estimate.**
Estimated CO_2e avoided	Project-level estimate from recovered quantity. **Project Estimate**, not a measured reduction.
Pickup completion rate	Share of pickup requests that reached COLLECTED.
Sustainability Score	Project-defined 0 100 metric (Section 11).

Dashboard charts: waste by category, monthly waste, recovery trend and resource-pathway distribution, drawn from database records with Chart.js.

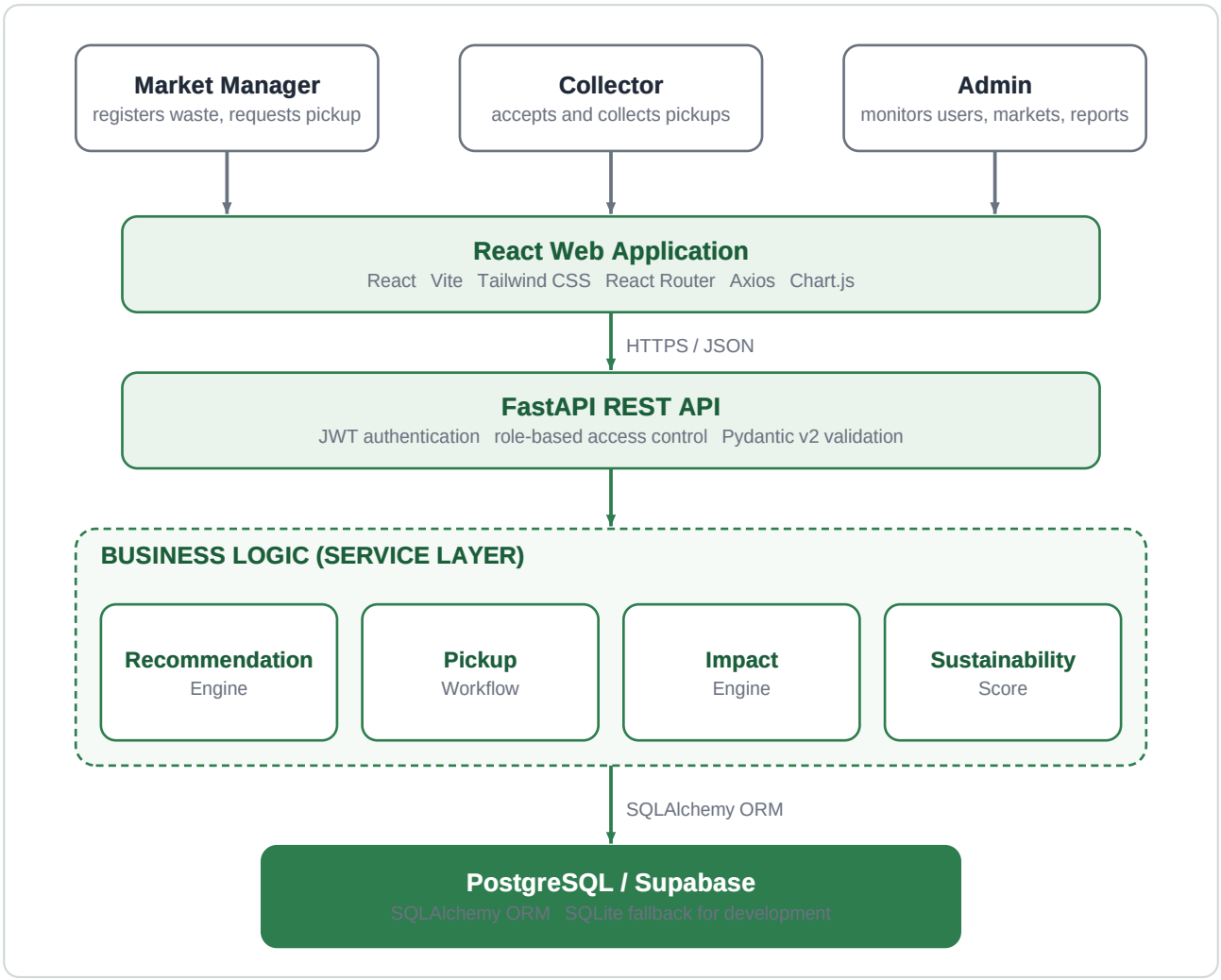
In the demonstration scenario, 150 kg of composted vegetable waste corresponds to 67.5 kg CO_2e avoided (**Project Estimate**), i.e. 0.45 kg CO_2e per kg under the prototype s demonstration setting. This is not independently scientifically verified.

13 Technology Stack

Layer	Technologies
Frontend	React, Vite, JavaScript, Tailwind CSS, Chart.js, React Router, Axios
Backend	Python, FastAPI, SQLAlchemy, Pydantic v2, JWT authentication, Bcrypt password hashing
Database	PostgreSQL / Supabase; SQLite development fallback
Testing	Pytest, HTTP/API testing

14 System Architecture

A three-tier architecture: one React application serves all three roles, a FastAPI service enforces authentication, authorization and business rules, and a relational database stores all records.



Authentication is handled in the FastAPI layer with JWT tokens and hashed passwords. Database credentials and secrets remain on the backend and are never exposed to the frontend.

15 User Roles

Role	Capabilities
Admin	User management market management waste registry pickup tracking reports platform monitoring
Market Manager	Dashboard waste registration recommendation viewing pickup requests impact monitoring sustainability score
Collector	Available pickups pickup acceptance collection updates verified collection history

Role-based access control ensures each role reaches only its own functions.

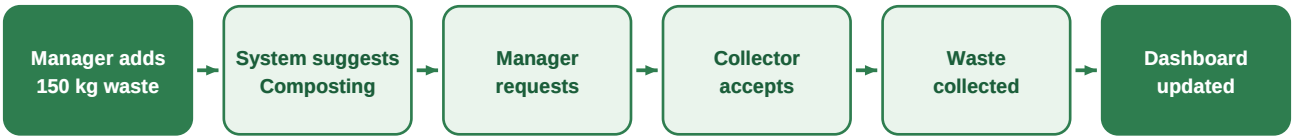
16 Sustainability & SDG Alignment

SDG	How BazarCycle BD relates to it
SDG 11 Sustainable Cities and Communities (primary)	Better coordination of market waste can contribute to cleaner and more sustainable urban environments. The platform organises how waste is registered, collected and tracked at the market level.
SDG 12 Responsible Consumption and Production (primary)	Supports waste segregation, recovery, recycling, composting and resource reuse by directing each category to a resource pathway instead of mixed disposal.
SDG 13 Climate Action (primary)	Better recovery of organic and recyclable waste has potential climate benefits. The prototype provides project-level CO_2e estimates (Project Estimate); it does not claim measured reductions.
SDG 8 Decent Work and Economic Growth (secondary)	Better organisation of local collection and recovery activities may support local economic activity. No employment creation is guaranteed or claimed.

These links describe potential contributions. BazarCycle BD is a prototype and does not claim to have achieved SDG targets.

17 Demonstration Scenario

The primary example used throughout this proposal:



Result	Demonstration value
Waste recovered	150 kg
Waste composted	150 kg
Recommended pathway	Composting
Estimated Resource Value	2,250 (150 kg 15/kg) Project Estimate
Estimated CO_2e avoided	67.5 kg Project Estimate

These values illustrate how the prototype works. They are not independently verified measurements and do not represent a real-market deployment.

18 Current Implementation Status

Status: Completed / Implemented Working Prototype. Not deployed at scale; no real-market usage is claimed.

Area	Implemented
Interface	React + Vite frontend; responsive mobile interface; accessibility improvements; animations and micro-interactions; loading and empty states; real market photography
Backend	FastAPI backend; JWT authentication; role-based access control
Data	PostgreSQL/Supabase support; SQLite development fallback
Core logic	Deterministic waste recommendation engine; BDT resource-value estimation; pickup workflow with concurrency protection
Monitoring	Impact dashboard; sustainability score

19 Testing & Validation

Check	Result
Backend tests (Pytest)	14/14 passed
Frontend production build	Passed successfully

The backend uses server-side validation: quantities must be greater than zero, and category and market are required. Validation covers the recommendation rules, value calculation and the pickup workflow.

Scope of validation. Testing confirms that the software behaves as designed. It does not validate environmental outcomes: no field trial, scientific measurement or real-market pilot has been carried out.

20 Expected Environmental Impact

- Encourages **segregation**, so organic and recyclable materials can be routed to composting or recycling.
- Gives markets a record of recovered waste, supporting reduced reliance on mixed disposal.
- Provides **Project Estimate** CO_2e indicators (for example 67.5 kg CO_2e for 150 kg composted in the demonstration) to compare scenarios within the prototype.
- Potential climate benefit through better organic and recyclable waste recovery; actual benefit would need field measurement.



Composting rings. Illustrative photograph supplied by the project team. It does not represent measured BazarCycle BD impact.

21 Expected Social Impact

- Cleaner market surroundings through organised collection and clearer responsibility.
- A transparent record of who requested, accepted and completed each pickup.
- Greater awareness among market managers of what their market generates.
- Clearer roles for collectors, supported by a verified collection history.

22 Expected Economic Impact

- Makes the potential **resource value** of waste visible (Project Estimate, 15/kg for the demonstration vegetable-waste valuation).
- May help organise local collection and recovery activity; no revenue, employment creation or cost saving is claimed.
- Estimated values vary by location, quality and market conditions and are not official market prices.

23 Limitations

- BazarCycle BD is a **prototype**: it has not been deployed at scale or used by real markets.
- Resource values and CO_2e figures are **Project Estimates**, not independently verified measurements.
- The Sustainability Score is project-defined and is not an official certification.
- The recommendation engine is rule-based and category-driven; it does not inspect actual waste composition.
- The platform coordinates collection but does not itself operate composting or recycling facilities.
- No partnerships, government support or market agreements are assumed in this proposal.
- Real-world adoption would depend on collector availability, facility access and market participation, which remain to be tested.

24 Future Scope

Direction	Description
Mobile application	Native or installable app for managers and collectors.
IoT smart bins	Sensor-assisted waste quantity capture.
Municipal integration	Possible coordination with city authorities, subject to agreement.
Real-time collection tracking	Live status of pickups in progress.
AI waste image classification	Photo-assisted categorization (not part of the current prototype).
Demand prediction	Forecasting waste generation and collection demand.
Advanced analytics	Deeper reporting and trend analysis.
Reward system	Incentives for consistent segregation and recovery.
Field pilot	Real-market trial to replace Project Estimates with measured data.

25 Conclusion

Market waste should not be treated only as waste. It can be tracked, redirected, recovered and measured as a resource.

BazarCycle BD responds to a coordination and monitoring gap with a working prototype: a rule-based recommendation engine, a protected pickup workflow, an impact dashboard and a project-defined Sustainability Score, tested with 14/14 backend tests and a successful frontend build.

Its figures are Project Estimates and its adoption is yet to be tested in the field. As a prototype, it offers a clear foundation for contributing to SDG 11, SDG 12 and SDG 13, and a practical basis for a future pilot.

Register Categorize Recommend Connect Collect Recover Measure

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